

Task 2 Pen & Paper

I am asked to find the solutions to the following differential equation:

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2} \quad (1)$$

Where n is some function of x , the spatial variable, and t the time variable, representing the number density of particles in a given the following conditions:

There is a periodic boundary condition:

$$n(x, t) = n(x + L, t) \quad (2)$$

The initial condition is as follows:

$$n(x, 0) = N\delta(x - x_0), \quad 0 \leq x < L \quad (3)$$

For some x_0 determining the starting position of our N particles, and some L as our bound.

The two different solutions I found are:

Through Fourier transform:

$$n(x, t) = \frac{N}{2\sqrt{\pi Dt}} \sum_m^{m \in \mathbb{Z}} e^{-\frac{(x-x_0+mL)^2}{4Dt}}$$

Through Fourier series

$$n(x, t) = \frac{N}{L} + \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(\cos \frac{2n\pi}{L} (x - x_0) \right)$$

The derivations are below.

To apply the boundary condition to the initial condition, we would usually have to describe it as a piecewise function. However, the dirac delta is a special type of function. Outside of $x = x_0$, $\delta(x - x_0) = 0$. So look at what happens if we take the sum of infinitely many dirac functions spaced L apart:

$$f(x) = N \sum_n^{n \in \mathbb{Z}} \delta(x - x_0 + nL)$$

In the original domain, from 0 to L , assuming $x_0 \in [0, L)$, then only the dirac function where $n = 0$ is nonzero in $[0, L)$. So in this interval, the following is true:

$$f(x) = N\delta(x - x_0) + N \sum_n^{n \in \mathbb{Z}, n \neq 0} \delta(x - x_0 + nL)$$
$$f(x) = N\delta(x - x_0)$$

Which satisfies our initial condition.

However, this function also satisfies the boundary condition:

$$f(x+L) = N \sum_{n \in \mathbb{Z}, n}^{n \in \mathbb{Z}, n} \delta(x - x_0 + nL + L)$$

$$f(x+L) = N \sum_{n \in \mathbb{Z}, n}^{n \in \mathbb{Z}, n} \delta(x - x_0 + (n+1)L)$$

Since n extends to all the integers, so does $n+1$, so ultimately this is still just an infinite sum of dirac deltas spaced L apart.

So $n(x, 0) = f(x)$ is our new initial condition that satisfies the original initial condition, and the boundary condition too. One thing that's even more useful is the fact that this is a linear homogeneous partial differential equation, so we can find the solution for just one of these dirac functions and then add them together.

To simplify the problem of finding this solution, we do a Fourier transform in the x domain on equation (1). This turns the second derivative of x into simply multiplication by ω . Let $\hat{n}(\omega, t) = \mathcal{F}(n)$:

$$\mathcal{F}\left(\frac{\partial n}{\partial t}\right) = \mathcal{F}\left(D \frac{\partial^2 n}{\partial x^2}\right) \quad (4)$$

$$\frac{\partial \hat{n}}{\partial t} = -D\omega^2 \hat{n} \quad (5)$$

Now this is a simple differential equation to solve, which we can do by substituting e^{kt} for $\hat{n}$ and solving for k , using the characteristic equation as well:

$$ke^{kt} = -D\omega^2 e^{kt} \quad (6)$$

$$k = -D\omega^2 \quad (7)$$

We arrive at:

$$\hat{n}(\omega, t) = \hat{a}(\omega) e^{-D\omega^2 t} \quad (8)$$

With an added function representing the constant of integration. This constant is only a constant with respect to t , **not necessarily** ω , so it is allowed to depend on ω .

If we plug $t = 0$ into (8), we can get the initial condition:

$$\hat{n}(\omega, 0) = \hat{a}(\omega) e^{-0D\omega^2} \quad (9)$$

$$\hat{n}(\omega, 0) = \hat{a}(\omega) \quad (10)$$

Now we need to transform the other two conditions. First, the initial condition:

$$\mathcal{F}(n(x, 0)) = \mathcal{F}(N\delta(x - x_0)) \quad (11)$$

$$\hat{n}(\omega, 0) = a(\omega) = \mathcal{F}(N\delta(x - x_0)) \quad (12)$$

We don't know the transform of the function on the right. Let's figure it out using the integral definition of the Fourier transform:

$$\mathcal{F}(N\delta(x - x_0)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} N\delta(x - x_0)e^{-i\omega x} dx \quad (13)$$

We can use an interesting property of the dirac delta function. If we take the integral of $\delta(x - a)f(x)$ where a is in the integration bounds, the result of the integral is simply $f(a)$. So we can plug in x_0 for x and continue:

$$\mathcal{F}(N\delta(x - x_0)) = \frac{1}{\sqrt{2\pi}} Ne^{-i\omega x_0} \quad (14)$$

Plugging into (12), we get:

$$a(\omega) = \frac{1}{\sqrt{2\pi}} Ne^{-i\omega x_0} \quad (15)$$

We can now plug this into (8) and get:

$$\hat{n}(\omega, t) = \frac{1}{\sqrt{2\pi}} Ne^{-i\omega x_0} e^{-D\omega^2 t} \quad (16)$$

$$\hat{n}(\omega, t) = \frac{1}{\sqrt{2\pi}} Ne^{-D\omega^2 t - i\omega x_0} \quad (17)$$

Now we take the inverse fourier transform of (17) to finally find our solution for $n(x, t)$:

$$\mathcal{F}^{-1}(\hat{n}) = \mathcal{F}^{-1} \left(\frac{1}{\sqrt{2\pi}} Ne^{-D\omega^2 t - i\omega x_0} \right) \quad (18)$$

We again use the integral definition:

$$n(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} Ne^{-D\omega^2 t - i\omega x_0} e^{i\omega x} d\omega \quad (19)$$

$$n(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} Ne^{-D\omega^2 t - i\omega x_0 + i\omega x} d\omega \quad (20)$$

$$n(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} Ne^{-D\omega^2 t + i\omega(x - x_0)} d\omega \quad (21)$$

Take note of the expression inside the exponent in the integral in (21):

$$-D\omega^2 t + i\omega(x - x_0) \quad (22)$$

$$-Dt \left(\omega^2 - \frac{1}{Dt} i\omega(x - x_0) \right) \quad (23)$$

Let's try to complete the square in ω so that it becomes easier to integrate:

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (24)$$

$$a = \omega \quad (25)$$

$$2ab = 2\omega b = \frac{1}{Dt} i\omega(x - x_0) \quad (26)$$

$$b = -\frac{1}{2Dt} i(x - x_0) \quad (27)$$

$$b^2 = -\frac{1}{4D^2 t^2} (x - x_0)^2 \quad (28)$$

Note that from (23), the terms inside the parenthesis correspond to a^2 and $2ab$. I will replace b , but not a since $a = \omega$.

$$-Dt \left(\omega^2 - \frac{1}{Dt} i\omega(x - x_0) \right) = -Dt(\omega^2 + 2\omega b) \quad (29)$$

$$= -Dt(\omega^2 + 2\omega b + b^2 - b^2) \quad (30)$$

$$= -Dt((\omega + b)^2 - b^2) \quad (31)$$

$$= -Dt(\omega + b)^2 + Dtb^2 \quad (32)$$

$$(33)$$

Replacing this inside our exponential from (21):

$$n(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} N e^{-Dt(\omega+b)^2 + Dtb^2} d\omega \quad (34)$$

$$n(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} N e^{-Dt(\omega+b)^2} e^{Dtb^2} d\omega \quad (35)$$

Since D , t , and b (see (27)) don't depend on ω , we can bring e^{Dtb^2} outside of the integral, as well as N :

$$n(x, t) = \frac{N}{2\pi} e^{Dtb^2} \int_{-\infty}^{\infty} e^{-Dt(\omega+b)^2} d\omega \quad (36)$$

Let's isolate the integral. We do a substitution: $u = \omega + b$, $du = d\omega$:

$$\int_{-\infty}^{\infty} e^{-Dt(u)^2} du \quad (37)$$

The bounds are tricky, but they are still infinity. If $\lim_{x \rightarrow \infty} x = \infty$, then $\lim_{x \rightarrow \infty} (x+b)$ for some constant b is still infinity, even if that constant is not real.

So with that, we do one final substitution. $v = u\sqrt{Dt}$, $dv = du\sqrt{Dt}$. The bounds stay infinite as before:

$$\frac{1}{\sqrt{Dt}} \int_{-\infty}^{\infty} e^{-v^2} dv \quad (38)$$

$$= \frac{\sqrt{\pi}}{\sqrt{Dt}} \quad (39)$$

We are left with a familiar integral, which simply evaluates to $\sqrt{\pi}$. Now we simply plug this into (36):

$$n(x, t) = \frac{N}{2\pi} e^{Dtb^2} \frac{\sqrt{\pi}}{\sqrt{Dt}} \quad (40)$$

Let's use (28) to bring back b^2 :

$$b^2 = -\frac{1}{4D^2t^2}(x - x_0)^2 \quad (41)$$

$$n(x, t) = \frac{N}{2\pi} e^{-Dt \frac{1}{4D^2t^2}(x-x_0)^2} \frac{\sqrt{\pi}}{\sqrt{Dt}} \quad (42)$$

$$n(x, t) = \frac{N}{2\sqrt{\pi Dt}} e^{-\frac{(x-x_0)^2}{4Dt}} \quad (43)$$

And we arrive at the solution to this equation for one dirac function. Now we sum infinitely many, spaced L apart, as specified before to get the full periodic solution:

$$n(x, t) = \frac{N}{2\sqrt{\pi Dt}} \sum_m^{m \in \mathbb{Z}} e^{-\frac{(x-x_0+mL)^2}{4Dt}}$$

While this solution is cool and accurate for small values of t , I foresee a large problem when it comes to its computation. I can only add together finitely many of those dirac terms, and no matter what, the t in the denominator in front of this formulation means this sum will decay eventually.

I will thus make another formulation using the fourier transform that works on a periodic boundary.

Method II: A different formulation

The heat equation affects sine and cosine waves predictably. If these waves have an angular frequency ω in the x domain, then they will decay according to $e^{-D\omega^2 t}$. We find the fourier series to break apart our initial condition into waves, multiply each one by this exponential, and add them back together to see how the function evolves over time, like so:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{2n\pi}{L} x + b_n \sin \frac{2n\pi}{L} x \right) \quad (44)$$

$$n(x, t) = a_0 + \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(a_n \cos \frac{2n\pi}{L} x + b_n \sin \frac{2n\pi}{L} x \right) \quad (45)$$

Where the function $f(x)$ is equal to our initial condition in the range of 0 to L , and is also periodic with a period of L .

All we need to do is find the fourier coefficients, beginning with a_0 :

$$a_0 = \frac{1}{L} \int_{-p}^p N \delta(x - x_0) dx \quad (46)$$

$$a_0 = \frac{N}{L} \quad (47)$$

Remember the useful properties of δ discussed on page 2.

Next is a_n :

$$a_n = \frac{2}{L} \int_{-p}^p N \delta(x - x_0) \cos \frac{2n\pi}{L} x dx \quad (48)$$

$$a_n = \frac{2N}{L} \cos \frac{2n\pi}{L} x_0 \quad (49)$$

And b_n :

$$b_n = \frac{2}{L} \int_{-p}^p N \delta(x - x_0) \sin \frac{2n\pi}{L} x dx \quad (50)$$

$$b_n = \frac{2N}{L} \sin \frac{2n\pi}{L} x_0 \quad (51)$$

Now we put it back into (45)

$$n(x, t) = \frac{N}{L} + \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(\frac{2N}{L} \cos \frac{2n\pi}{L} x_0 \cos \frac{2n\pi}{L} x + \frac{2N}{L} \sin \frac{2n\pi}{L} x_0 \sin \frac{2n\pi}{L} x \right) \quad (52)$$

$$n(x, t) = \frac{N}{L} + \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(\cos \frac{2n\pi}{L} x_0 \cos \frac{2n\pi}{L} x + \sin \frac{2n\pi}{L} x_0 \sin \frac{2n\pi}{L} x \right) \quad (53)$$

$$n(x, t) = \frac{N}{L} + \frac{2N}{L} \sum_{n=1}^{\infty} e^{-D(\frac{2n\pi}{L})^2 t} \left(\cos \frac{2n\pi}{L} (x - x_0) \right) \quad (54)$$

With this solution, we now have a bunch of cosine terms that decay to 0, while we have a constant term equal to N/L , which is exactly what it should be: the density of particles.

The problem with this one is that it may be a bit more unstable closer to $t = 0$ and thus requires more high frequency components there. It seems the fourier formulation is good for larger D and t , or once it has already spread out, while the gaussian formulation is good for smaller t when everything is still concentrated.

With this, QED.